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import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import make_blobs
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA
from sklearn.metrics import silhouette_score
print("--- Part 1: K-Means Clustering ---")
X, y = make_blobs(
    n_samples=300,
    centers=3,
    cluster_std=0.60,
    random_state=42
)
df_kmeans = pd.DataFrame(
    X,
    columns=["Feature_1", "Feature_2"]
)
print("\nOriginal K-Means Dataset Head:")
print(df_kmeans.head())
wcss = []
for k in range(1, 11):

    kmeans = KMeans(
        n_clusters=k,
        init="k-means++",
        max_iter=300,
        n_init=10,
        random_state=42
    )
    kmeans.fit(X)
    wcss.append(kmeans.inertia_)
plt.figure(figsize=(10, 6))
plt.plot(
    range(1, 11),
    wcss,
    marker="o",
    linestyle="--"
)
plt.title("Elbow Method for Optimal K")
plt.xlabel("Number of Clusters (K)")
plt.ylabel("WCSS")
plt.grid(True)
plt.show()
optimal_k = 3
kmeans = KMeans(
    n_clusters=optimal_k,
    init="k-means++",
    max_iter=300,
    n_init=10,
    random_state=42
)

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clusters = kmeans.fit_predict(X)
df_kmeans["Cluster"] = clusters
print("\nK-Means Clustered Data:")
print(df_kmeans.head())
plt.figure(figsize=(10, 8))
sns.scatterplot(
    x="Feature_1",
    y="Feature_2",
    hue="Cluster",
    data=df_kmeans,
    s=100
)
plt.scatter(
    kmeans.cluster_centers[:, 0],
    kmeans.cluster_centers[:, 1],
    s=300,
    marker="X",
    label="Centroids"
)
plt.title("K-Means Clustering with K=3")
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.legend()
plt.grid(True)
plt.show()
silhouette_avg = silhouette_score(
    X,
    clusters
)
print(
    "\nSilhouette Score for K-Means:",
    round(silhouette_avg, 3)
)
print("\n--- Part 2: Dimensionality Reduction with PCA ---")
X_pca, y_pca = make_blobs(
    n_samples=500,
    n_features=4,
    centers=4,
    cluster_std=1.0,
    random_state=25
)
df_pca = pd.DataFrame(
    X_pca,
    columns=[
        "Feature_1",
        "Feature_2",
        "Feature_3",
        "Feature_4"
    ]
)
df_pca["True_Cluster"] = y_pca
print("\nOriginal PCA Dataset Head:")
print(df_pca.head())
print("\nOriginal PCA Dataset Shape:")
print(df_pca.shape)
scaler = StandardScaler()
X_pca_scaled = scaler.fit_transform(X_pca)
pca = PCA(n_components=2)
principal_components = pca.fit_transform(

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    X_pca_scaled
)
df_pca_result = pd.DataFrame(
    principal_components,
    columns=[
        "Principal_Component_1",
        "Principal_Component_2"
    ]
)
df_pca_result["True_Cluster"] = y_pca
explained_variance = pca.explained_variance_ratio_

print("\nPrincipal Components Head:")
print(df_pca_result.head())

print("\nExplained Variance Ratio:")
print(explained_variance)

print(
    "\nTotal Explained Variance by 2 PCs:",
    round(explained_variance.sum(), 3)
)
plt.figure(figsize=(10, 8))

sns.scatterplot(
    x="Principal_Component_1",
    y="Principal_Component_2",
    hue="True_Cluster",
    data=df_pca_result,
    s=100
)
plt.title("PCA - Dimensionality Reduction to 2 Components")
plt.xlabel(
    f"PC1 ({explained_variance[0] * 100:.2f}%)
)
plt.ylabel(
    f"PC2 ({explained_variance[1] * 100:.2f}%)
)
plt.grid(True)
plt.show()
kmeans_pca = KMeans(
    n_clusters=4,
    init="k-means++",
    max_iter=300,
    n_init=10,
    random_state=42
)
clusters_pca = kmeans_pca.fit_predict(
    principal_components
)
df_pca_result["KMeans_Cluster_on_PCA"] = clusters_pca
plt.figure(figsize=(10, 8))
sns.scatterplot(
    x="Principal_Component_1",
    y="Principal_Component_2",
    hue="KMeans_Cluster_on_PCA",
    data=df_pca_result,
    s=100

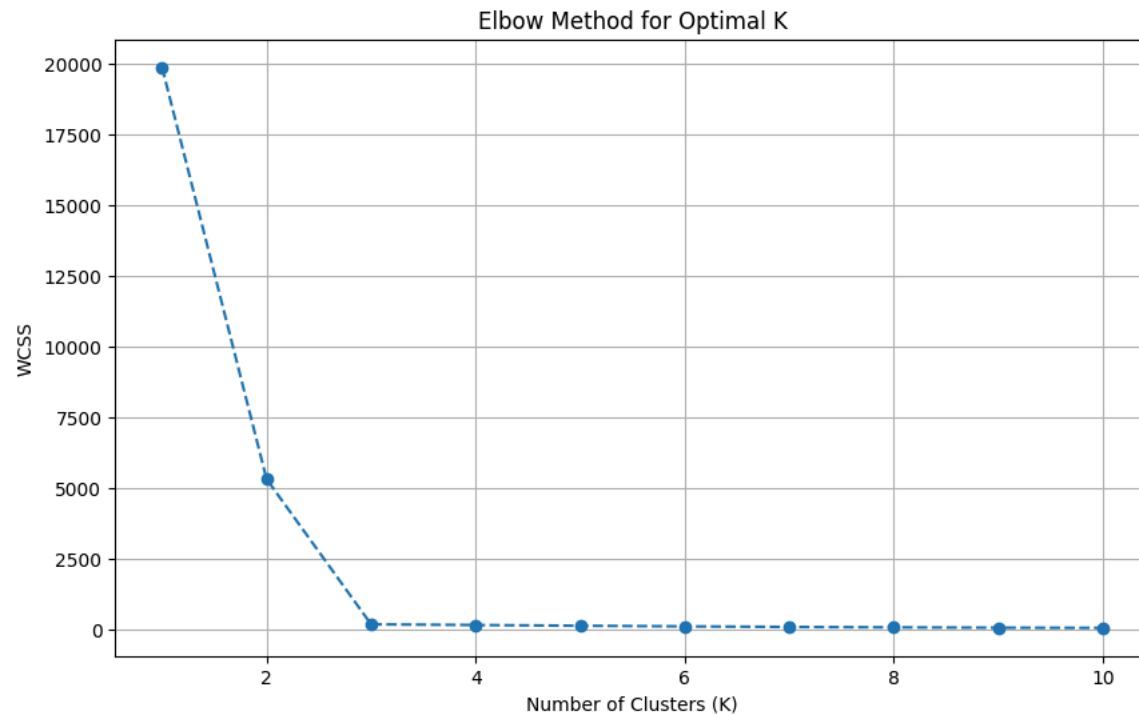
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)
plt.scatter(
    kmeans_pca.cluster_centers[:, 0],
    kmeans_pca.cluster_centers[:, 1],
    s=300,
    marker="X",
    label="Centroids"
)
plt.title("K-Means Clustering on PCA-Reduced Data")
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.legend()
plt.grid(True)
plt.show()
silhouette_avg_pca = silhouette_score(
    principal_components,
    clusters_pca
)
print(
    "\nSilhouette Score for K-Means on PCA-Reduced Data:",
    round(silhouette_avg_pca, 3)
)
print("\n=====")
print("EXPERIMENT 5 COMPLETED SUCCESSFULLY")
print("=====")
```


--- Part 1: K-Means Clustering ---

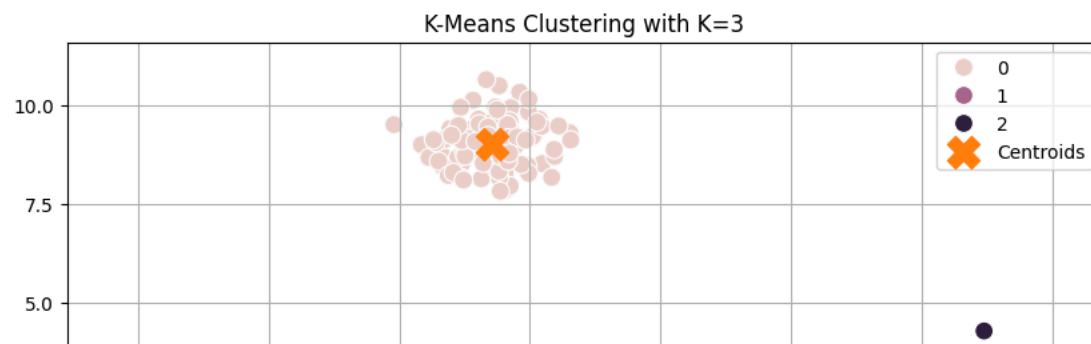
Original K-Means Dataset Head:

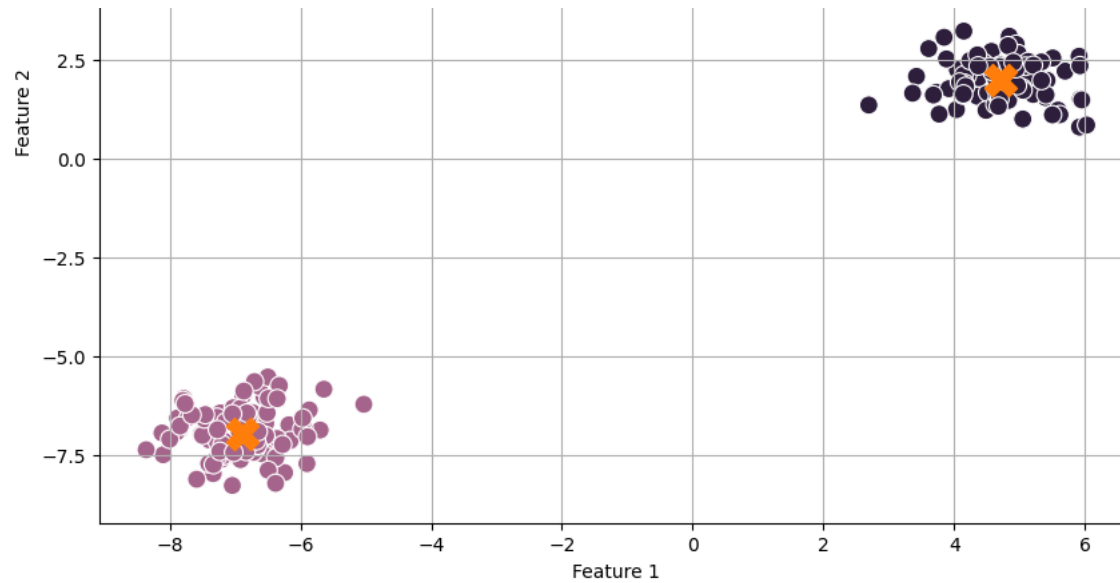
	Feature_1	Feature_2
0	-7.155244	-7.390016
1	-7.395875	-7.110843
2	-2.015671	8.281780
3	4.509270	2.632436
4	-8.102502	-7.484961



K-Means Clustered Data:

	Feature_1	Feature_2	Cluster
0	-7.155244	-7.390016	1
1	-7.395875	-7.110843	1
2	-2.015671	8.281780	0
3	4.509270	2.632436	2
4	-8.102502	-7.484961	1





Silhouette Score for K-Means: 0.908

--- Part 2: Dimensionality Reduction with PCA ---

Original PCA Dataset Head:

	Feature_1	Feature_2	Feature_3	Feature_4	True_Cluster
0	-0.638667	1.110057	-6.400722	-0.204990	3
1	-2.951556	-7.657445	3.844794	0.903589	1
2	-0.253177	2.125103	-7.869801	0.559678	3
3	-2.151209	3.401400	-5.734930	0.965230	3
4	-2.347519	-7.230467	3.478891	-0.443440	1

Original PCA Dataset Shape:

(500, 5)

Principal Components Head:

	Principal_Component_1	Principal_Component_2	True_Cluster
0	0.455305	1.623917	3
1	-2.705622	0.375012	1
2	0.810234	1.966926	3
3	0.427139	2.149626	3
4	-2.407508	0.099250	1

Explained Variance Ratio:

[0.57208431 0.33622342]

Total Explained Variance by 2 PCs: 0.908

PCA - Dimensionality Reduction to 2 Components

